

## Analysis PhD Exam, June 1996

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*Remark. In the sequel the word measure means positive, countably additive measure, unless otherwise stated.*

*Instructions. Do all of the following problems.*

1. Let  $f_n : \mathbb{R} \rightarrow \mathbb{R}$  be a sequence of Lebesgue-measurable functions which converges pointwise to a function  $g$ . Prove, directly from the definition of measurability, that  $g$  is measurable.
2. For  $f \in L^1(\mathbb{R})$ , let  $f_t$  denote the translation of  $f$  by  $t$ , i.e.  $f_t(x) := f(x - t)$ . Show that the map  $t \mapsto f_t$  is a continuous map from  $\mathbb{R}$  to  $L^1(\mathbb{R})$ .
3. Let  $\mu$  denote the Lebesgue measure on  $\mathbb{R}$ . Let  $T : \mathbb{R} \rightarrow \mathbb{R}$  be defined by  $T(x) = x^3 - x$ . Define a Borel measure  $\nu$  setting  $\nu(A) := \mu(T^{-1}A)$ . Compute the Radon–Nikodym derivative  $\frac{d\nu}{d\mu}$ .
4. Let  $K_n \subset [0, 1]$  be a Cantor set of Lebesgue measure larger than  $1 - \frac{1}{n}$ . Let  $V := \bigcup_{n=1}^{\infty} K_n$ . Prove that, for any null set  $N$ , the set  $V \setminus N$  is not a  $G_\delta$  set. [Hint: Baire Category Theorem]
5. Let  $\mu$  and  $\nu$  be  $\sigma$ -finite measures on the measurable space  $(X, \mathcal{B})$ . Prove, from first principles, that  $\nu$  can be written as  $\nu = \nu_{ac} + \nu_{sing}$ , with  $\nu_{ac} \ll \mu$  and  $\nu_{sing} \perp \mu$ .
6. Let  $x_n$  be a sequence in a Banach space  $B$ . Assume that the sequence  $x_n$  converges weakly to  $x$ .
  - (a) Show that  $x$  is the only weak limit of the sequence  $x_n$ .
  - (b) Show that  $\sup_n \|x_n\| < \infty$ . [Hint for (b): Consider the  $x_n$  as elements of the double dual  $B^{**}$ .]
7. Let  $K : [0, 2\pi] \rightarrow \mathbb{R}$  be defined by  $K(x) = \frac{1}{4} \sin(x)$ . Show that for every  $h \in L^p([0, 2\pi])$ , there exists a unique solution  $f$  to the equation

$$f + f * K = h.$$

If  $h$  is  $C^\infty$ , is it true that the solution  $f$  is  $C^\infty$ ? Justify your answer.

8. A distribution  $T$  is called harmonic if  $\Delta T = 0$ , where  $\Delta = \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$ . Show that if  $T$  is a harmonic tempered distribution, then  $T$  is a polynomial. [Hint: a distribution with support  $\{0\}$  is a finite sum of multiples of Dirac's  $\delta$  and its derivatives.]